

## Figure 2 | A signal is usable only inside its validated setting

| Setting                     | Risk signal<br>Spearman $\rho$ | Magnitude<br>Spearman $\rho$ | Interpretation      |
|-----------------------------|--------------------------------|------------------------------|---------------------|
| E189 random pair            | 0.37–0.41                      | 0.24                         | both positive       |
| E189 unseen column          | 0.21–0.25                      | −0.09                        | risk signal only    |
| E189 unseen row             | −0.10–−0.01                    | 0.10*                        | risk negative       |
| E189 double unseen          | −0.35–−0.24                    | −0.08                        | ABSTAIN             |
| E190 cross-study K562       | 0.42                           | 0.42                         | similar             |
| E192 cross-study RPE1       | 0.30 (CI×0)                    | 0.34 (CI×0)                  | ABSTAIN             |
| E199 unseen genes · K562    | 0.395 ✓                        | 0.096 (CI×0)                 | disagreement valid  |
| E200 context holdout · K562 | 0.424                          | 0.880 ✓                      | magnitude dominates |
| E158 strict unseen          | undefined                      | undefined                    | PRESCRIBE saturated |
| E201 four contexts          | SEALED                         | SEALED                       | evaluation pending  |

\* point estimate only; CI crossing 0 means the system must abstain. E201 is sealed, not a result.